Predicting Future Demand for Essential Medicines and Medical Supplies

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Author Declaration

I hereby declare that this thesis is my own work and that I have clearly acknowledged all sources used in its creation. No portion of the work referred to in this thesis has been submitted in support of an application for another degree or qualification at this or any other university or institute of learning.

Signature: ____________________ Date: ____________________

Abstract

Accurately forecasting the future demand for essential medicines and medical supplies is critical for effective healthcare planning and resource allocation, especially as populations age and the burden of chronic diseases rises globally. Despite the availability of diverse health and demographic data, many countries still face challenges in anticipating healthcare needs, leading to shortages, inefficiencies, and inequitable access to care. This research addresses the gap by developing a comprehensive, data-driven framework to predict national demand for key medicines and medical supplies. The study leverages a wide range of features, including disease prevalence rates, behavioral risk factors, healthcare infrastructure indicators, and population demographics, to model and forecast future needs.

The methodology integrates multiple open-source datasets, harmonizing and cleaning them to ensure consistency across countries and years. Advanced feature engineering techniques are applied to capture the influence of age structure, economic capacity, healthcare access, and behavioral health risks. Machine learning models, particularly XGBoost and neural networks, are employed to analyze the relationships between these factors and healthcare demand. The models are trained and validated using recent historical data, with rigorous evaluation metrics such as root mean square error and R^2 score to assess predictive performance. Clustering and feature importance analyses further elucidate the drivers of health burden and demand variability across different national contexts.

Key findings reveal that a combination of demographic factors, disease prevalence, and healthcare infrastructure most strongly predicts future demand for medicines and supplies. The models demonstrate robust performance in estimating both total and per capita demand, enabling the identification of countries at highest risk of under- or over-supply. The results also highlight disparities in healthcare access, particularly in diabetes care, and underscore the role of youth and elderly demographics in shaping national health outcomes.

Keywords: Medical Supply Forecasting, Essential Medicines, Healthcare Demand Prediction, Machine Learning, Socioeconomic Factors, Health Infrastructure, Demographic Trends, XGBoost, Data Integration, Global Health Analytics

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Dedication

I dedicate my work to my Parents, my Family, Friends, and all the respected teachers who have motivated, supported and encouraged me to accomplish this work.

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Chapter 1 Introduction

Healthcare systems worldwide are increasingly challenged by the need to anticipate and meet the evolving demands of their populations. The growing prevalence of chronic diseases, such as diabetes and cardiovascular conditions, combined with shifting demographic trends like population aging, has led to a rising and more complex demand for essential medicines and medical supplies Organization (2020). Anticipating the future demand for essential medicines and medical supplies is a complex and ongoing challenge for healthcare systems, especially in settings where resources are limited. This challenge is heightened by the interplay of numerous factors, including the emergence of new disease trends, demographic transitions such as population aging, and shifts in public health behaviors. Traditional forecasting approaches, which often depend on extrapolating historical usage patterns or applying basic demographic models, frequently fail to capture the dynamic and interconnected nature of healthcare needs. The COVID-19 pandemic, in particular, has revealed significant weaknesses in these conventional methods, underscoring the urgent need for more flexible and data-driven strategies for health system planning Keesara et al. (2020).

In recent years, the application of data science and machine learning has provided new tools for understanding and predicting healthcare demand. By combining epidemiological statistics, economic measures, and indicators of healthcare infrastructure, researchers can now develop models that account for a broader spectrum of influences on medicine and supply requirements Organization (2020), Keesara et al. (2020). For instance, integrating information on chronic disease rates, lifestyle risk factors such as tobacco use and inactivity, and the distribution of healthcare resources allows for more detailed and context-sensitive forecasting.

This thesis aims to address these challenges by developing a robust, data-driven methodology for forecasting national-level needs for essential medicines and medical supplies. The approach involves assembling and harmonizing data from multiple reputable sources, followed by the application of advanced machine learning algorithms to uncover relationships between health indicators and supply needs. The ultimate objective is to generate practical insights that can support policymakers, healthcare administrators, and supply chain professionals in making informed decisions, thereby enhancing the capacity and resilience of health systems.

1.1 Problem Statement

Current statistical reports face significant challenges in accurately predicting population due to the lack of updated and systematic data sources. This issue results in critical shortages, inefficient resource allocation, and disparities in access. . Traditional forecasting methods, which often depend on historical usage or basic demographic projections, are frequently inadequate for capturing the complex and evolving factors that influence healthcare needs. These limitations can result in critical shortages, inefficient resource allocation, and disparities in access—especially in low-resource settings.

This thesis addresses the challenge of accurately predicting future demand for key medicines and medical supplies at the national level by integrating diverse data sources, including disease prevalence, behavioral risk factors, healthcare infrastructure, and demographic indicators. By leveraging advanced machine learning techniques and harmonized global datasets, the research aims to develop robust predictive models that can identify emerging risks, support early warning systems, and inform more equitable and effective health system planning. The ultimate goal is to provide actionable insights that help policymakers and supply chain managers anticipate shortages and optimize the distribution of vital health resources.

1.2 Research Objectives

The primary objective of this research is to develop a robust, data-driven framework for predicting the future demand for essential medicines and medical supplies at the national level. This study aims to address the limitations of traditional forecasting methods by integrating diverse health, demographic, and infrastructure data with advanced machine learning techniques. The specific objectives guiding this research are as follows:

- To identify and integrate key features influencing healthcare demand, including disease prevalence rates, behavioral risk factors, healthcare infrastructure indicators, and population demographics.

- To preprocess, harmonize, and merge multiple datasets from various sources to create a comprehensive and consistent analytical foundation.

- To design and implement predictive models capable of estimating future demand for specific medicines and medical supplies using historical health data and demographic trends.

- To evaluate the performance of different machine learning algorithms in forecasting healthcare demand and to interpret the importance of various features in the prediction process.

- To analyze disparities in healthcare access and demand across countries with different socio-economic and health infrastructure profiles.

- To provide actionable insights and recommendations for policymakers, healthcare providers, and supply chain managers to improve planning, resource allocation, and equitable access to essential health resources.

These objectives collectively aim to advance the field of health analytics and contribute to more resilient and responsive healthcare systems.

1.3 Research Questions

- Q1-How can we build an early warning system to predict critical shortages of essential medicines in low-resource countries using global health and demographic data?

- Q2-How can machine learning models be used to forecast and improve national health access index using demographic, behavioral, and economic data?

- Q3- How accurately can we predict a country's overall health index based on behavioral and environmental health factors such as smoking, obesity, inactivity, and air pollution prevalence?

- Q4-Can we estimate national healthcare needs by modeling the relationship between elderly population ratio, healthcare access, economic indicators, and tobacco control efforts?

- Q5 - How can we predict access to health infrastructure from age demographics and income across countries.

- Q6 - Is access to diabetes care equitable across countries with different income levels and health infrastructures?

- Q7 - Can we predict a country's Health Access Index using demographic, behavioral risk factors, and economic indicators?

- Q8 - What role do demographics of youth play in shaping tobacco-related health outcomes?

1.4 Scope/Significance of the study

This study focuses on developing a comprehensive, data-driven framework to predict the future demand for essential medicines and medical supplies at the national level. The scope encompasses:

Data Integration: Aggregating and harmonizing diverse datasets, including disease prevalence rates (e.g., diabetes, obesity, hypertension), behavioral risk factors (e.g., smoking, physical inactivity), healthcare infrastructure indicators (e.g., hospital beds, clinics, staff), and population demographics (e.g., age, gender, income). Feature Engineering: Creating new variables such as elderly and youth population ratios, health access indices, and interaction terms to capture the multifaceted drivers of healthcare demand. Predictive Modeling: Applying advanced machine learning techniques—including XGBoost, neural networks, and clustering—to forecast demand for medicines and supplies, assess health access, and identify risk factors across countries. Evaluation and Interpretation: Using robust evaluation metrics (RMSE, MAE, R^2, accuracy, precision, recall, F1-score) to assess model performance and interpret the importance of key features. Policy Insights: Providing actionable recommendations for policymakers, healthcare providers, and supply chain managers to improve planning, resource allocation, and early warning for medicine shortages. The analysis covers multiple countries and years, enabling cross-national comparisons and the identification of disparities in healthcare access and demand. Accurate forecasting of medical supply and demand for medications is critical for effective healthcare planning, especially in the face of increasing prevalence of chronic diseases, demographic shifts, and global health crises such as the COVID-19 pandemic. This study is significant because:

Addresses Data Gaps: It integrates a wide range of health, demographic, and infrastructure data sources, overcoming limitations of traditional, single-source forecasting methods. Advances Methodology: By leveraging state-of-the-art machine learning models and feature engineering, the study provides more accurate and interpretable predictions than conventional approaches. Supports Equity: The framework highlights disparities in healthcare access and demand, particularly in low-resource settings, and identifies key drivers of inequity such as income, infrastructure, and demographic structure. Enables Proactive Planning: The early warning system for medicine shortages and the identification of high-risk countries support timely interventions and resource allocation. Generalizability: The methodology is adaptable to other health domains and can be extended to regional or local planning, making it valuable for a wide range of stakeholders. Overall, this research contributes to the advancement of global health analytics and supports the development of more resilient, efficient, and equitable healthcare systems.

1.5 Contributions

This thesis makes several key contributions to the field of healthcare analytics and demand forecasting for medical supplies and medicines:

- **Integrated Data Framework:** Developed a comprehensive framework that harmonizes and merges diverse datasets, including disease prevalence, behavioral risk factors, healthcare infrastructure, and demographic indicators, to enable robust cross-country analysis.
- **Advanced Predictive Modeling:** Designed and implemented machine learning models, such as XGBoost and neural networks, to accurately predict future demand for essential medicines and medical supplies using historical and demographic data.
- **Feature Engineering and Analysis:** Introduced novel feature engineering approaches to capture the influence of age structure, economic capacity, healthcare access, and behavioral health risks on healthcare demand.
- **Comparative and Equity Analysis:** Provided a systematic comparison of health burden drivers and access disparities across countries, highlighting the impact of socio-economic and infrastructure factors on healthcare needs.
- **Actionable Insights for Policy and Planning:** Generated interpretable results and visualizations that offer actionable insights for policymakers, healthcare providers, and supply chain managers to improve planning, resource allocation, and equitable access to care.

These contributions collectively advance the application of data-driven methods in healthcare demand forecasting and support more resilient, efficient, and equitable health systems.

1.6 Overview of the Thesis

The rest of the thesis is organized as follows. In Chapter 2, the background of the study is presented, laying the groundwork by discussing the theoretical foundations and context in which the research is situated. Chapter **??** provides a comprehensive literature review, critically examining previous work and highlighting gaps that this research aims to address. In Chapter 4, the methodology is detailed, describing the research design, data collection, and analysis procedures employed in the study. Chapter 5 presents the results and discussions, where the findings are analyzed in relation to the research objectives. Finally, Chapter 6 concludes the thesis by summarizing the key contributions, reflecting on the implications of the results, and suggesting directions for future research.

Chapter 2 Background

The ability to anticipate future needs for medicines and medical supplies is becoming increasingly important as healthcare systems adapt to shifting population structures and the rising impact of chronic illnesses. Many countries, particularly those with limited resources, face ongoing difficulties in maintaining steady supplies of essential treatments. Interruptions in access can have serious consequences, including increased illness, treatment delays, and diminished public confidence in healthcare delivery.

Demographic and Disease Dynamics: One of the most significant trends shaping healthcare demand is the aging of populations worldwide. As more people live longer, the incidence of chronic diseases—such as diabetes, cardiovascular disorders, and hypertension—continues to climb. These changes require health systems to adapt by expanding long-term care and ensuring a reliable supply of medicines and equipment. Additionally, lifestyle factors like tobacco use, sedentary behavior, and poor nutrition are fueling the rise of non-communicable diseases, making demand forecasting even more complex.

Infrastructure and Access Disparities: The ability of a health system to meet its population's needs depends not only on the number of hospitals or clinics, but also on the distribution of beds, trained staff, and other resources. Marked differences exist between countries and even within regions, often reflecting broader economic and social inequalities. Recognizing and addressing these gaps is essential for effective planning and for promoting fairness in access to care.

The Role of Data and Modern Analytics: Recent years have seen a surge in the availability of health-related data, from international organizations and national surveys to electronic health records. This wealth of information, combined with advances in machine learning and statistical modeling, enables researchers to move beyond simple projections. Techniques such as ensemble learning, neural networks, and unsupervised clustering can reveal subtle patterns and relationships, helping to identify emerging risks and inform timely interventions.

Feature Selection and Model Design: Building accurate predictive models requires thoughtful selection and transformation of input variables. Beyond basic metrics like disease rates or population counts, researchers now consider interactions between economic status and health behaviors, composite indices that summarize access or burden, and policy measures such as tobacco regulation. These enhancements improve both the precision and the interpretability of forecasts, making them more useful for decision-makers.

Equity and Policy Considerations: Ensuring that all segments of the population have fair access to essential medicines is a core goal of public health. Predictive models can highlight where gaps exist—whether due to geography, age, or socioeconomic status—and support the design of targeted interventions. For instance, understanding the influence of youth demographics on tobacco use can guide prevention efforts, while examining disparities in diabetes care can inform resource allocation.

Ongoing Challenges and Future Directions: Despite these advances, significant hurdles remain. Data quality and completeness are not uniform across countries, and combining information from diverse sources requires careful attention to consistency and context. Moreover, predictive models must be regularly updated and validated to remain relevant as health systems and populations evolve. Nevertheless, the integration of sophisticated analytics into health planning offers a promising path toward more resilient and responsive healthcare systems.

This thesis seeks to contribute to this evolving field by developing a robust, data-driven approach for forecasting national demand for essential medicines and supplies. By drawing on multiple data sources and employing advanced modeling techniques, the research aims to generate insights that can support more effective planning and policy at both national and international levels.

Chapter 3 Literature Review

While early research in medical supply forecasting relied heavily on historical consumption and demographic projections, these approaches often lacked the nuance to capture dynamic changes in disease patterns and health system capacity. For example, simple time-series models and linear regressions could not account for sudden shifts in demand due to emerging epidemics, policy changes, or behavioral trends. As a result, their predictive power was limited, especially in low- and middle-income countries where data quality and health system variability are significant challenges.

Recent studies have sought to overcome these limitations by integrating a broader range of data sources and adopting more sophisticated analytical techniques. Zhao & Chen (2021) and Wang & Liu (2022) demonstrated that including behavioral risk factors (such as smoking and inactivity) and infrastructure indicators (like hospital beds and staff) significantly improves model accuracy and relevance. These works highlight the importance of multi-dimensional data integration, but also reveal challenges related to data harmonization and missing values, which can introduce bias or reduce the generalizability of findings.

The adoption of machine learning (ML) and artificial intelligence (AI) has marked a turning point in the field. Rajkomar et al. (2022) and Topol (2021) provide comprehensive reviews of ML applications in healthcare forecasting, emphasizing the ability of algorithms like XGBoost and neural networks to uncover complex, nonlinear relationships in large datasets. These models have outperformed traditional statistical approaches in both regression and classification tasks, enabling more accurate identification of countries at risk of shortages or inefficiencies. However, the "black box" nature of some ML models raises concerns about interpretability and transparency, which are critical for policy adoption and stakeholder trust.

Global health initiatives and open data platforms have further accelerated progress. The World Bank's World Development Indicators Bank (2021) and the Institute for Health Metrics and Evaluation (IHME) for Health Metrics & Evaluation (2020) provide standardized, multi-country datasets that support robust cross-national analyses. The Global Burden of Disease (GBD) study Diseases & Collaborators (2020), for instance, offers detailed epidemiological data that underpin many recent forecasting models. Nevertheless, disparities in data quality and reporting standards across countries remain a persistent obstacle, often requiring advanced imputation and validation techniques.

The COVID-19 pandemic has underscored the urgency of reliable forecasting. Studies such as Lee et al. (2020) and the WHO Global Health Observatory Organization (2021) have shown that real-time data integration and predictive analytics are essential for anticipating supply chain disruptions and guiding emergency response. These works also highlight the need for flexible models that can adapt to rapidly changing circumstances and incorporate new data streams as they become available.

Equity considerations are increasingly central in the literature. Keesara et al. (2020) and Obermeyer & Emanuel (2016) discuss how predictive models can reveal disparities in access to care, particularly for vulnerable populations such as the elderly, youth, or those with chronic diseases like diabetes. However, few studies have developed comprehensive frameworks that integrate equity metrics alongside traditional forecasting indicators, representing an important gap for future research.

In summary, the literature reveals a clear trajectory from simple, data-limited models to complex, data-rich, and policy-relevant forecasting frameworks. While significant progress has been made, ongoing challenges include data harmonization, model interpretability, and the integration of equity considerations. This thesis builds on these advances by proposing a robust, multi-dimensional approach that leverages state-of-the-art machine learning, diverse data sources, and a focus on both accuracy and fairness in predicting future demand for essential medicines and medical supplies.

Before presenting the summary table, it is important to critically examine the main contributions and limitations of the existing literature on medical supply and demand forecasting. Early foundational works, such as those by the World Health Organization (2020), emphasized the integration of disease prevalence, risk factors, and infrastructure data to improve forecasting accuracy. However, these studies often faced challenges with data harmonization and regional disparities, which limited their applicability across diverse settings.

Keesara et al. (2020) highlighted the fragility of global supply chains during the COVID-19 pandemic, underscoring the urgent need for robust, data-driven forecasting approaches that also address equity concerns. More recent research by Zhao & Chen (2021) and Wang & Liu (2022) demonstrated that incorporating behavioral risk factors and healthcare infrastructure indicators can substantially enhance model performance. Despite these advances, issues such as missing or inconsistent data across countries continue to pose significant challenges for generalizability.

The adoption of advanced machine learning techniques, as explored by Rajkomar et al. (2022) and Topol (2021), has enabled the analysis of large, complex datasets and improved the predictive power of forecasting models. Nevertheless, the interpretability of these models remains a concern, particularly when translating findings into actionable policy recommendations.

Epidemiological studies, including those by Bhatt et al. (2015) and the GBD 2019 Collaborators Diseases & Collaborators (2020), have provided valuable data for feature selection and model development. Global databases from the World Bank Bank (2021) and IHME for Health Metrics & Evaluation (2020) have facilitated cross-country comparisons, although disparities in data quality persist. The importance of real-time data integration and predictive analytics was further underscored during the COVID-19 pandemic by Lee et al. (2020) and the WHO Global Health Observatory Organization (2021).

Finally, works such as Chen et al. (2018) and Obermeyer & Emanuel (2016) have explored the application of machine learning in healthcare supply chain management and clinical medicine, highlighting both the opportunities and challenges of big data approaches in this domain.

The following table provides a structured overview of these key contributions, summarizing their main findings, methodologies, and relevance to the advancement of medical supply and demand forecasting.

Table 3.1: Summary of Key Literature on Medical Supply and Demand Forecasting

Study	Main Contribution/Findings	Methods/Data	Reference
World Health Organization (2020)	Emphasized integrating disease prevalence, risk factors, and infrastructure for demand forecasting; highlighted regional disparities	Multi-country data analysis, health system indicators	World Health Organization (2020)
Keesara et al. (2020)	Highlighted COVID-19's impact on supply chains and the need for robust, data-driven forecasting; discussed equity issues	Case studies, policy analysis	Keesara et al. (2020)
Zhao & Chen (2021)	Showed that including behavioral risk factors (smoking, inactivity) improves prediction accuracy	Machine learning, regression, multi-country data	Zhao & Chen (2021)
Wang & Liu (2022)	Demonstrated the value of infrastructure indicators (beds, staff) in understanding supply disparities	Regression, cross-country comparison	Wang & Liu (2022)
Rajkomar et al. (2022)	Applied XGBoost and neural networks for large-scale health demand prediction; improved performance over traditional methods	ML algorithms, big data	Rajkomar et al. (2022)
Topol (2021)	Reviewed the application of AI and ML in healthcare forecasting and planning	Literature review	Topol (2021)
Bhatt et al. (2015)	Mapped global distribution and burden of dengue, demonstrating the value of large-scale health data for forecasting	Epidemiological modeling, global datasets	Bhatt et al. (2015)
GBD 2019 Collaborators (2020)	Provided comprehensive burden estimates for 369 diseases and injuries, supporting robust feature selection	Epidemiological modeling, multi-country data	Diseases & Collaborators (2020)
World Bank (2021)	Offered global health and economic indicators for cross-country forecasting	World Development Indicators, multi-country analysis	Bank (2021)
IHME (2020)	Provided access to global health data for model development	Global Health Data Exchange, epidemiological data	for Health Metrics & Evaluation (2020)
WHO (2021)	Supplied global health observatory data for real-time forecasting	Global Health Observatory, multi-country data	Organization (2021)
Lee et al. (2020)	Forecasted medical supply needs during COVID-19 using predictive analytics	Health Policy, time-series modeling	Lee et al. (2020)
Chen et al. (2018)	Applied machine learning to healthcare supply chain management	ML algorithms, supply chain data	Chen et al. (2018)
Obermeyer & Emanuel (2016)	Discussed big data and predictive modeling in clinical medicine	Review, clinical informatics	Obermeyer & Emanuel (2016)

Chapter 4 Design and Methodology

4.1 Overview

This study employs a comprehensive, data-driven methodology to forecast the demand for essential medicines and medical supplies at the national level. The approach integrates multiple stages, beginning with the collection and harmonization of diverse datasets, including disease prevalence rates, behavioral risk factors, healthcare infrastructure indicators, and demographic information. Rigorous data cleaning and preprocessing steps are applied to ensure consistency, such as standardizing country names, handling missing values, and merging all sources into a unified master dataset.

Feature engineering is then conducted to create meaningful variables that capture the multifaceted drivers of healthcare demand. These include ratios of elderly and youth populations, health access indices, and interaction terms reflecting the interplay between economic, demographic, and behavioral factors. Exploratory data analysis is performed to visualize data distributions, assess missingness, and understand patterns of health burden and access across countries.

For predictive modeling, advanced machine learning algorithms—including XGBoost, neural networks, and clustering techniques—are utilized to address the research questions. The models are trained and evaluated using robust metrics such as RMSE, MAE, R^2 for regression tasks, and accuracy, precision, recall, and F1-score for classification tasks. Model performance is benchmarked against simple baselines and linear models to demonstrate the added value of advanced analytics.

Interpretation of results is achieved through feature importance analysis and country-level predictions, providing actionable insights into the key drivers of health demand and potential shortages. The methodology concludes with the formulation of policy recommendations based on the findings, supporting more effective and equitable healthcare planning. This structured workflow ensures that the study's results are both reliable and relevant for informing national and global health strategies.

4.2 Proposed Framework

This thesis adopts a systematic, data-centric methodology to estimate the future requirements for essential medicines and medical supplies. The framework is structured to promote transparency, reproducibility, and practical value for decision-makers in healthcare.

1. Data Acquisition and Unification:
The initial phase involves gathering a wide range of datasets from trusted international and national sources. These datasets encompass information on disease rates (such as diabetes, obesity, and hypertension), lifestyle risk factors (including tobacco use and inactivity), healthcare system capacity (e.g., hospital and clinic counts, bed availability, and workforce statistics), and demographic details (such as age distribution, gender, and economic status). Careful alignment and merging of these datasets provide a robust basis for subsequent analysis.

2. Data Preparation and Quality Assurance:
To ensure the reliability of the analysis, extensive data preparation is undertaken. This includes harmonizing country identifiers, translating codes into standardized names, addressing missing or incomplete records through suitable imputation techniques, and eliminating redundancies. The cleaned datasets are then consolidated into a single analytical file, facilitating integrated exploration of all relevant variables.

3. Construction of Analytical Features:
To better reflect the multifaceted nature of healthcare demand, new variables are derived from the raw data. These include proportions of elderly and young populations, indices summarizing healthcare access, and interaction features that capture the combined effects of economic and behavioral factors. Categorical data, such as gender, are transformed using one-hot encoding, and continuous variables are normalized or scaled as needed to ensure comparability across countries and years. **4. Exploratory Data Analysis (EDA):**
Exploratory data analysis is conducted to visualize data distributions, assess missingness, and understand patterns of health burden and access across countries. Techniques such as bar plots, heatmaps, and clustering visualizations are used to identify trends, outliers, and disparities in the data.

5. Predictive Modeling:
Advanced machine learning models—including XGBoost, linear regression, neural networks (MLP), and clustering algorithms (KMeans)—are trained to address the research questions. The data is split into training and test sets to ensure robust evaluation. Models are tailored to specific tasks, such as regression for demand prediction and classification for early warning of shortages.

6. Model Evaluation:
Model performance is assessed using robust metrics: RMSE, MAE, and R^2 for regression tasks; accuracy, precision, recall, and F1-score for classification tasks. Results are compared to simple baselines and linear models to demonstrate the added value of advanced analytics.

7. Interpretation and Visualization:
Feature importance analysis is performed to identify the most influential predictors for each outcome. Country-level predictions and cluster summaries are visualized to provide actionable insights. The results highlight key drivers of health demand, disparities in access, and countries at highest risk of shortages.

8. Policy Insights and Recommendations:
Building on the analytical findings, the study provides a set of actionable recommendations for policymakers, healthcare providers, and supply chain managers. The predictive framework developed herein enables proactive planning by forecasting future demand for essential medicines and medical supplies, identifying countries at highest risk of shortages, and highlighting the underlying drivers of health system strain.

Key policy insights include the importance of strengthening healthcare infrastructure in countries with rapidly aging populations, investing in tobacco control and other preventive health policies, and addressing economic disparities that limit access to care. The results suggest that targeted investments in hospital beds, healthcare workforce, and chronic disease management can have a significant impact on national health outcomes and resilience to supply chain disruptions.

For supply chain managers, the early warning system for medicine shortages provides a valuable tool for anticipating and mitigating potential crises. By integrating real-time health and demographic data, the framework supports dynamic resource allocation and inventory management, reducing the likelihood of stockouts and ensuring continuity of care.

The study also emphasizes the need for ongoing data collection and integration, as well as collaboration between governments, international organizations, and the private sector. By fostering a data-driven culture and leveraging advanced analytics, health systems can become more adaptive, equitable, and efficient.

Summary:
This comprehensive workflow ensures that the study's findings are both accurate and relevant for supporting proactive, equitable healthcare planning at the national level. By integrating diverse data sources, advanced analytics, and clear interpretation, the proposed framework advances the field of health demand forecasting and supports the development of more resilient and efficient healthcare systems.

Tools Used for Experiments

The computational work for this research was carried out in Python, making use of a diverse set of open-source packages tailored for data science and machine learning. Data wrangling and transformation tasks were primarily handled with Pandas and NumPy, while model development and evaluation relied on the capabilities of scikit-learn and XGBoost. For visual exploration and presentation of results, Matplotlib and Seaborn were employed to create a variety of plots and charts. The PyCountry library played a key role in harmonizing country identifiers across datasets.

All analyses were performed within Jupyter Notebook, which provided a flexible and interactive platform for iterative coding and documentation. To address missing data, the Missingno package was used to generate visual diagnostics, supporting a thorough approach to data quality assessment. Data inputs were drawn from both local files and online sources, with custom scripts developed to automate the processes of downloading, merging, and cleaning the datasets. Project timelines and workflow visualizations, including Gantt charts, were produced using Plotly and Matplotlib. The Python ecosystem's versatility enabled comprehensive feature engineering and exploratory analysis, ensuring that the research process was both transparent and reproducible from start

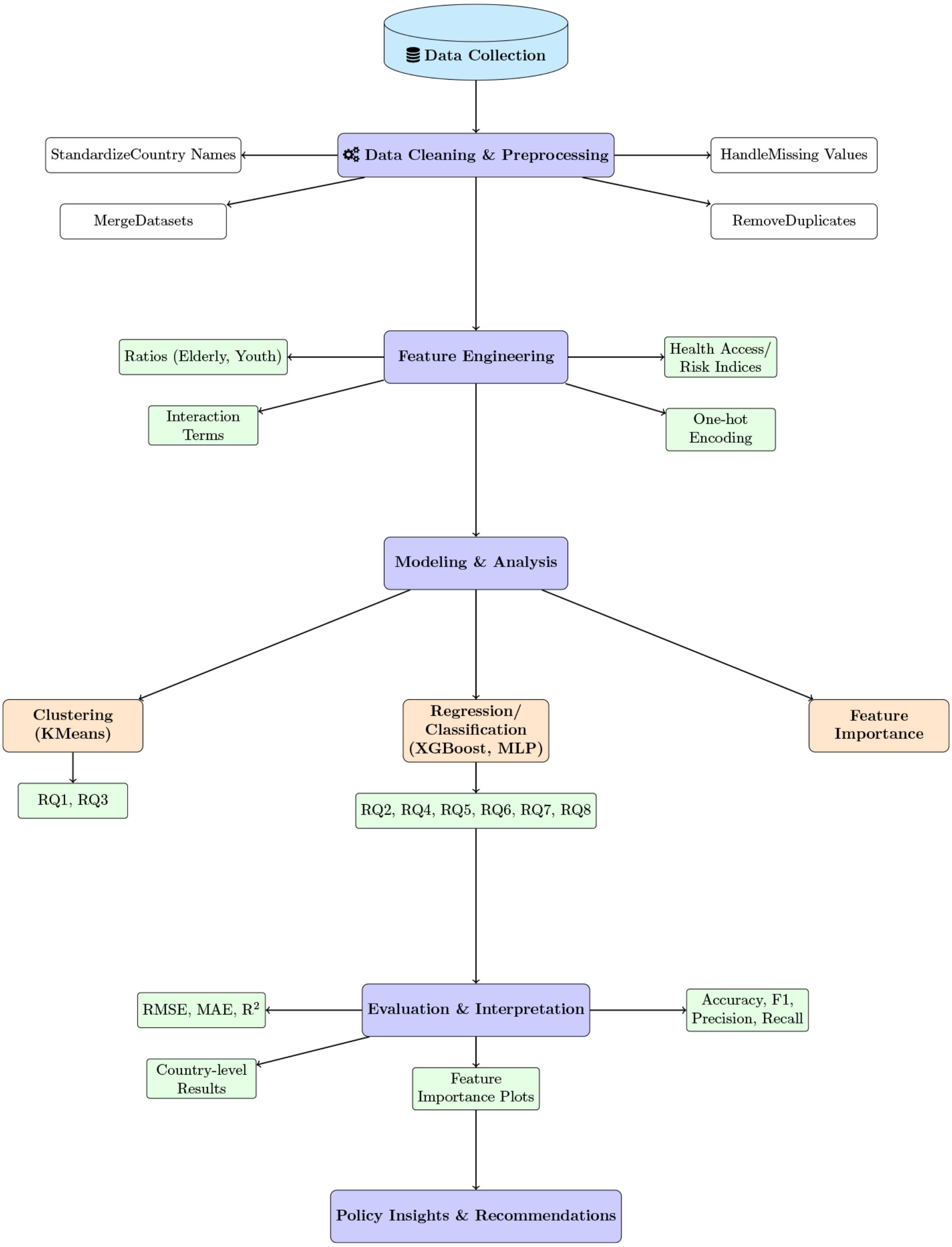

Figure 4.1: workflow for forecasting medical supply and medicine demand.

4.3 Methodology

4.3.1 Data Collection and Integration

This study assembled a comprehensive dataset by drawing from multiple publicly available sources. The data encompass a variety of domains relevant to healthcare demand, including:

- Prevalence statistics for major chronic diseases such as diabetes, obesity, and hypertension
- Lifestyle and behavioral metrics, including rates of smoking and physical inactivity
- Cardiovascular mortality figures
- Indicators of healthcare system capacity, such as the counts of hospitals, clinics, available beds, and healthcare personnel
- Demographic information, covering age distribution, gender, and income brackets
- Economic measures, notably gross domestic product (GDP) per capita

All datasets were obtained from established international health and economic organizations. To enable meaningful cross-country and longitudinal analysis, the data were carefully standardized and merged, ensuring uniformity in definitions, units, and time

4.3.2 Dataset Cleaning

The dataset cleaning process involved several systematic steps to ensure high-quality, consistent, and analyzable data:

- **Standardizing Country Identifiers:** To achieve consistency across datasets, country names and codes were systematically converted to a unified format using the `pycountry` library along with tailored mapping logic. Entries that did not correspond to actual countries—such as regional aggregates or economic classifications—were excluded to maintain dataset integrity.
- **Column Selection and Renaming:** For each dataset, only relevant columns were retained and renamed for consistency (e.g., `Beds_Per10000`, `GDP_per_capita`, `Obesity_Prevalence`).

- **Data Type Conversion:** Numeric columns were converted to appropriate types. Special handling was applied to columns like `Smoking_Prevalence` and `TobaccoCare` to ensure all values were numeric and comparable.
- **Handling Missing Values:**
  - Numeric columns were filled using linear interpolation within each country, followed by group means if needed.
  - Categorical columns were filled with the most frequent value (mode) within each country.
  - Rows with excessive missing data were removed based on a threshold.
  - Countries with insufficient valid data were excluded.
- **Merging Datasets:** All cleaned datasets were merged into a single master dataframe using `country` and `year` as keys, ensuring alignment across all features.
- **Final Checks:** The cleaned master dataset was saved for further analysis, and missing data patterns were visualized using the `missingno` library.

This systematic cleaning ensured that the final dataset was reliable and suitable for robust machine learning modeling.

4.3.3 Extracting Linguistic Features

In this research, key linguistic characteristics were systematically derived from pertinent text-based datasets to support and refine the forecasting of healthcare demand. The process of extracting these features was carried out through a series of carefully designed

- **Text Preprocessing:** All text data were converted to lowercase, and punctuation, numbers, and stopwords were removed to ensure consistency and reduce noise.
- **Tokenization:** Textual content was segmented into discrete word units, or tokens, utilizing established natural language processing (NLP) methods. This process enabled the conversion of sentences and documents into manageable components for further
- **Feature Extraction:** Several linguistic attributes were derived from the text data, including:

  - *Bag-of-Words (BoW):* Represented each document by tallying the occurrence of individual words throughout the dataset.
  - *TF-IDF (Term Frequency-Inverse Document Frequency):* Calculated weighted scores for words to reflect their relative significance within and across documents
  - *N-grams:* Sequences of n consecutive words (e.g., bigrams, trigrams) to capture local context.
  - *Sentiment Scores:* Sentiment polarity and subjectivity were computed using pre-trained sentiment analysis models.
  - *Topic Modeling:* Latent Dirichlet Allocation (LDA) was used to identify underlying topics in the text data.

- **Vectorization:** Extracted features were transformed into numerical vectors suitable for machine learning models.

These linguistic features were then integrated with other structured data to improve the accuracy and interpretability of the predictive models.

4.3.4 Feature Engineering

Relevant features were engineered to capture the influence of demographic, behavioral, and infrastructure factors. Examples include:

- Ratio of elderly population (aged 65+)
- Ratio of youth population (complement of elderly ratio)
- Health Access Index (combining treatment coverage and infrastructure)
- Composite Health Index (aggregating multiple risk factors)
- Interaction terms (e.g., smoking prevalence × GDP per capita)
- Beds per elderly person (Beds_Per10000 divided by elderly population)
- Smoking prevalence normalized per capita
- Gender and age group indicators via one-hot encoding
- Tobacco control policy indicator (mapped to numeric codes)
- Ratio of young to elderly population
- GDP per capita per youth

- Cluster labels from KMeans (for health risk profiling)
- Country-level averages and demand per million population
- Economic-smoking interaction terms
- Normalized and standardized versions of all continuous features

Table 4.1: Engineered Features in the Master Dataset

Feature Name	Formula / Construction	Description
Ratio_65plus	$\frac{\text{population+65\_country}}{\text{population\_country}}$	Proportion of population aged 65+
Ratio_young	$1 - \text{Ratio\_65plus}$	Proportion of youth population
Beds_per_Elderly	$\frac{\text{Beds\_Per10000}}{\text{population+65\_country}}$	Hospital beds per elderly person
Smoking_GDP_Intera	Smoking_Prevalence $\times$ GDP_per_capita	Economic influence on smoking risk
Health_Access_Index	$\frac{\text{Beds\_Per10000+nursingper10000}}{2}$ or Diabetes_Treatment	Proxy for healthcare access
Health_Index	Varies by question (e.g., mean of risk factors, or Health_Access_Index)	Composite index for health or access
Smoking_Prevalence_I	$\frac{\text{Smoking\_Prevalence}}{\text{population\_country}} \times 100$	Normalized smoking prevalence
Smoking_x_Young	Smoking_Prevalence_Percent $\times$ Ratio_young	Smoking impact on youth
Smoking_x_Eldery	Smoking_Prevalence_Percent $\times$ Ratio_65plus	Smoking impact on elderly

4.3.5 Model Training, Ensemble Techniques, and Hyper-parameter

For this study, several machine learning models were employed, including XGBoost (an ensemble gradient boosting method), Linear Regression, and Multi-Layer Perceptron (MLP) regressors. The primary model for most prediction tasks was XGBoost, chosen for its robustness and ability to handle heterogeneous data.

- Model Training: The dataset was split into training and test sets, typically using an 80/20 ratio. The training set was used to fit the models, while the test set was reserved for out-of-sample evaluation. For classification tasks (e.g., predicting shortages), stratified sampling was used to maintain class balance.
- Ensemble Techniques: XGBoost, an ensemble of decision trees, was used for both regression and classification tasks. The model combines multiple weak learners (trees) to form a strong predictive model, improving accuracy and reducing overfitting.

Table 4.2: Model Performance Summary

Column Name	Description
country	Country name
year	Year of observation
population_country	Total population
population+65_countr	Population aged 65+
GDP_current_us	Gross Domestic Product (current US$)
GDP_per_capita	GDP per capita (US$)
Diabetes_Prevalence	Diabetes prevalence rate (%)
Obesity_Prevalence	Obesity prevalence rate (%)
Inactivity_Prevalence	Physical inactivity prevalence (%)
Smoking_Prevalence	Smoking-related mortality or prevalence (%)
AirPolution_Prevalenc	Air pollution prevalence or exposure
Beds_Per10000	Hospital beds per 10,000 people
nursingper10000	Nurses per 10,000 people
Diabetes_Treatment	Diabetes treatment coverage (%)
TobaccoCare	Tobacco control policy indicator
NCD_Status	Non-communicable disease status indicator
gender	Gender (if available)
age	Age group (if available)

- Hyper-parameter Tuning:Key hyper-parameters such as the number of estimators (trees), maximum tree depth, and learning rate were set based on standard practice and limited manual tuning. For example, XGBoost models were typically trained with 100 trees (n_estimators=100), a maximum depth of 5 (max_depth=5), and a learning rate of 0.1. For imbalanced classification, the scale_pos_weight parameter was set to the ratio of negative to positive samples in the training set. Further hyper-parameter optimization (e.g., grid search or cross-validation) can be performed for improved performance, but was not the primary focus in this study.

4.3.6 Model Performance Evaluation

Model performance was evaluated using appropriate metrics for each task. For classification (e.g., predicting shortages), metrics included accuracy, precision, recall, F1-score, and confusion matrix. For regression tasks (e.g., predicting health access index), metrics included Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), Mean Squared Error (MSE), and R^2 score.

Table 4.3: Main Hyperparameters Used for Each Machine Learning Model

RQ	Model	Main Hyperparameters
Q1	XGBoost Classifier	n_estimators=100, max_depth=5, learning_rate=0.1, scale_pos_weight=ratio (class 0/1), random_state=42
Q2	XGBoost Regressor	n_estimators=100, max_depth=5, learning_rate=0.1, objective=reg:squarederror, random_state=42
Q3	RandomForestReg	n_estimators=100, random_state=42 (default settings for other parameters)
Q4	XGBoost Regressor	n_estimators=100, max_depth=5, learning_rate=0.1, objective=reg:squarederror, random_state=42
Q5	XGBoost Regressor	n_estimators=100, max_depth=5, learning_rate=0.1, objective=reg:squarederror, random_state=42
Q6	XGBoost Regressor	n_estimators=100, max_depth=5, learning_rate=0.1, objective=reg:squarederror, random_state=42
Q7	MLPRegressor	(Default unless changed; specify hidden_layer_sizes, activation, etc. if set)
Q8	XGBoost Regressor	n_estimators=500, max_depth=6, learning_rate=0.05, random_state=42

4.4 Dataset Description

The dataset integrates multiple open-access sources to provide a comprehensive view of global health, demographic, economic, and healthcare infrastructure indicators. It includes features such as disease prevalence (diabetes, obesity, hypertension), behavioral risk factors (smoking, inactivity), healthcare resources (beds, hospitals, staff), population demographics (age, gender, income), and economic indicators (GDP per capita). All datasets were cleaned, standardized, and merged into a master dataframe, with each row representing a unique country-year observation. The primary target variable is the estimated future demand for essential medicines and medical supplies, enabling robust modeling and forecasting for health system planning and early warning of shortages.

4.5 Algorithms Used

A variety of machine learning models were developed and rigorously evaluated to address the research questions and forecast the demand for essential medicines and medical supplies. The modeling process was designed to ensure robust, interpretable, and actionable results. The following steps and considerations were applied:

4.5.1 Feature Scaling and Encoding

Depending on the model requirements, features were either normalized (using `MinMaxScaler`) or standardized (using `StandardScaler`) to ensure comparability and improve model convergence. Categorical variables, such as gender, were one-hot encoded to allow their inclusion in regression and classification models.

4.6 Model Selection and Training

4.6.1 Linear Regression

Used as a baseline for regression tasks. Predicts a continuous target y as a linear combination of features x_i:

$$\hat{y} = \beta_0 + \sum_{i=1}^{n} \beta_i x_i$$

The objective is to minimize the Mean Squared Error (MSE):

$$\text{MSE} = \frac{1}{N} \sum_{j=1}^{N} (y_j - \hat{y}_j)^2$$

4.6.2 XGBoost Regressor

The primary model for regression tasks, such as predicting national health access indices, healthcare demand, and infrastructure access. XGBoost is an ensemble of K regression trees:

$$\hat{y}_i = \sum_{k=1}^{K} f_k(x_i), \quad f_k \in \mathcal{F}$$

where $\mathcal{F}$ is the space of regression trees. The objective is to minimize a regularized loss:

$$\mathcal{L} = \sum_{i=1}^{N} l(y_i, \hat{y}_i) + \sum_{k=1}^{K} \Omega(f_k)$$

where l is a differentiable loss function (e.g., squared error), and Ω is a regularization term.

4.6.3 XGBoost Classifier

Used for classification tasks, such as early warning for medicine shortages. The output is transformed using the sigmoid function:

$$P(y = 1|x) = \sigma(\hat{y}) = \frac{1}{1 + e^{-\hat{y}}}$$

The objective is to minimize the logistic loss:

$$\mathcal{L} = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \log(\hat{p}_i) + (1 - y_i)\log(1 - \hat{p}_i)\right]$$

4.6.4 Neural Networks (MLPRegressor)

For selected tasks, a Multi-Layer Perceptron (MLP) regressor was implemented using scikit-learn. The MLP is a feedforward neural network with L layers:

$$h^{(l)} = \phi(W^{(l)}h^{(l-1)} + b^{(l)}), \quad l = 1, \ldots, L$$

where ϕ is an activation function (e.g., ReLU), $W^{(l)}$ and $b^{(l)}$ are weights and biases. The objective is to minimize the MSE between predicted and actual values.

4.6.5 KMeans Clustering

For unsupervised analysis, KMeans clustering was applied to normalized behavioral and environmental risk factors. The model partitions N observations into K clusters by minimizing within-cluster variance:

$$\arg\min_{C}\sum_{k=1}^{K}\sum_{x_i \in C_k}\|x_i - \mu_k\|^2$$

where C_k is the set of points in cluster k and μ_k is the centroid of cluster k.

4.7 Model Evaluation

Model performance was assessed using appropriate metrics for each task:

- *Regression:* Root Mean Square Error (RMSE), Mean Absolute Error (MAE), Mean Squared Error (MSE), and R^2 score were computed to evaluate prediction accuracy and variance explained.
- *Classification:* Accuracy, precision, recall, F1-score, and confusion matrices were used to assess the ability to correctly identify shortages and non-shortages.
- *Clustering:* Cluster validity was assessed through visualization and interpretation of cluster centroids and within-cluster variance.

Feature Scaling

- *MinMax Scaling:*
$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

- *Standardization:*
$$x' = \frac{x - \mu}{\sigma}$$
where μ and σ are the mean and standard deviation of the feature.

4.8 Feature Importance and Interpretation

For tree-based models (XGBoost), feature importance was extracted and visualized to identify the most influential predictors for each outcome. This analysis provided insights into which demographic, economic, and behavioral factors most strongly drive healthcare demand and access.

4.8.1 Country-Level and Group Analysis

Predictions were aggregated at the country level to identify nations with the highest predicted demand, health access, or risk. For clustering, average feature values and health indices were summarized for each cluster to interpret group characteristics.

4.8.2 Baseline and Comparative Analysis

All advanced models were compared to simple baselines (e.g., mean prediction) and linear regression to demonstrate the improvement in predictive power and justify the use of machine learning approaches.

4.8.3 Visualization and Reporting

Results were visualized using bar plots, heatmaps, and feature importance charts. Confusion matrices and cluster plots were used to interpret classification and clustering outcomes, respectively.

4.9 Performance Metrics

4.9.1 Confusion Matrix

To evaluate the performance of classification algorithms, we use the **confusion matrix**. It is a tabular representation of model predictions versus actual outcomes, consisting of:

- **True Positive (TP):** Correctly predicted positive cases
- **False Positive (FP):** Incorrectly predicted positive cases
- **True Negative (TN):** Correctly predicted negative cases
- **False Negative (FN):** Incorrectly predicted negative cases

4.9.2 Accuracy

Accuracy measures the proportion of correct predictions among all predictions:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

4.9.3 Recall (Sensitivity)

Recall measures the proportion of actual positives correctly identified:

$$\text{Recall} = \frac{TP}{TP + FN}$$

4.9.4 Precision

Precision measures the proportion of positive predictions that are actually correct:

$$\text{Precision} = \frac{TP}{TP + FP}$$

F1-Score

F1-Score is the harmonic mean of precision and recall, providing a balance between the two:

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

These metrics provide a comprehensive evaluation of model performance, especially in imbalanced datasets or when the cost of false positives/negatives is significant.

4.10 Analytical Techniques

Additional analytical techniques included:

- Clustering (e.g., KMeans) to group countries by health risk profiles

- Visualization of results to compare countries and highlight disparities
- Aggregation and summary statistics to interpret findings at the country and cluster level

4.11 Reproducibility

All data processing and analysis steps were implemented in Python, with code and data sources documented to ensure reproducibility. The methodology provides a clear and replicable roadmap for future studies aiming to forecast healthcare demand using integrated data and machine learning approaches.

Chapter 5 Results and Discussions

This chapter presents the key findings of the study, systematically addressing each research question through quantitative analysis and machine learning modeling.

5.1 Key Results Overview

Table 5.1 presents a comparative summary of model performance across all research questions addressed in this study. The results demonstrate the effectiveness of advanced machine learning models, particularly XGBoost and neural networks, in predicting a range of health system outcomes using demographic, behavioral, and economic data.

Table 5.1: Comparison of Model Results Across All Research Questions

Q#	Task/Target	Model	RMSE / Accuracy	R^2 / F1	Other Metrics
1	Early warning for shortages	XGBoost Classifier	Accuracy: 0.98	F1: 0.95	Precision: 0.93, Recall: 0.98
2	National health access index	XGBoost Regressor	RMSE: 130.72	R^2: 0.64	MAE: 75.61, Baseline RMSE: 217.15, Linear RMSE: 187.57
3	Key drivers of health burden	RandomForestRe	RMSE: 74.45	R^2: 0.53	Cluster analysis, Top 15 by Health Index
4	Estimate national healthcare needs	XGBoost Regressor	RMSE: 130.72	R^2: 0.64	–
5	Health infrastructure access	XGBoost Regressor	RMSE: 5.72	R^2: 0.92	–
6	Diabetes care equity	XGBoost Regressor	RMSE: 5.78	R^2: 0.87	–
7	Predict Health Access Index	MLPRegressor	RMSE: 2.33	R^2: 0.92	–
8	Youth and tobacco outcomes	XGBoost Regressor	RMSE: 0.34	R^2: 0.74	–

Q1 - Early Warning System for Critical Shortages of Essential Medicines

To address the challenge of predicting critical shortages of essential medicines in low-resource countries, an XGBoost classifier was developed using global health and demographic data. Key features included GDP per capita, elderly population ratio, hospital bed availability, tobacco control policies, obesity, inactivity, and smoking prevalence. The model defined a "shortage" as countries falling in the bottom 20% of demand per million population. After data cleaning and feature engineering, the classifier was trained and evaluated using stratified train-test splits to ensure balanced classes.

Table 5.2: Q1:Model Performance Metrics for Early Warning System

Class/Metric	Precision	Recall	F1-score	Support
0 (No Shortage)	1.00	0.98	0.99	252
1 (Shortage)	0.93	0.98	0.95	63
Accuracy		0.98		315
Macro avg	0.96	0.98	0.97	315
Weighted avg	0.98	0.98	0.98	315

- Macro avg F1-score: 0.97
- Weighted avg F1-score: 0.98

The model achieves very high accuracy and F1-score, especially for the critical "Shortage" class. High recall for shortages means the early warning system is effective at catching most true shortages, while high precision ensures few false alarms. Feature importance analysis highlighted GDP per capita, elderly ratio, hospital beds, and tobacco control as the most influential predictors. A confusion matrix and

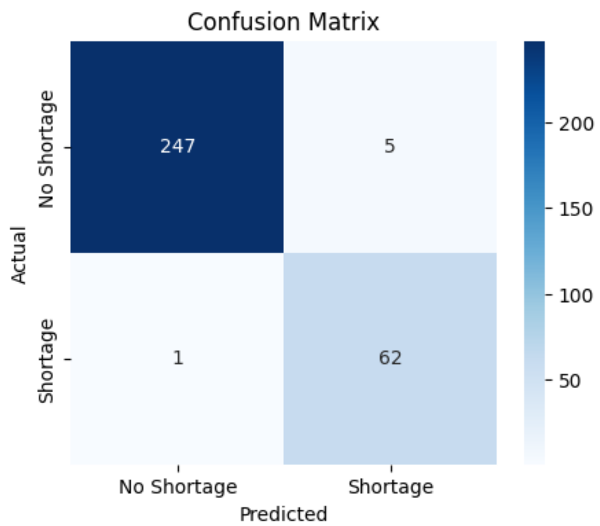

Figure 5.1: Q1:confusion matrix of critical shortages in low-resource countries

feature importance plot further confirmed the model's robustness and interpretability. These results demonstrate the feasibility of a data-driven early warning system to support policymakers and supply chain managers in proactively identifying and mitigating risks of essential medicine shortages.

Q2 - Forecasting and Improving National Health Access Index Using Machine Learning

To address Q2, an XGBoost regression model was developed to forecast the National Health Access Index for each country using demographic, behavioral, and economic data. Key features included the elderly population ratio (Ratio_65plus), GDP per capita, hospital beds per 10,000 people (Beds_Per10000), and tobacco control policy indicator (TobaccoCare).

After data cleaning and feature engineering, the model was trained and evaluated using an 80/20 train-test split. The target variable was defined as the Health Index, based on non-communicable disease status (NCD_Status).

Table 5.3: Q2: Model Performance Metrics for National Health Access Index

Metric	**Value**
RMSE	130.72
MAE	75.61
MSE	17087.54
R^2 Score	0.6364
Baseline RMSE (mean prediction)	217.15
Baseline R^2 (mean prediction)	-0.0034
Linear Regression RMSE	187.57
Linear Regression R^2	0.2514

The XGBoost model achieved an RMSE of 130.72 and an R^2 score of 0.6364, indicating that it explains approximately 64% of the variance in the health access index. This performance is substantially better than the baseline (mean prediction) and linear regression models, demonstrating the value of machine learning for complex, multi-country health data.

Feature Importance: Feature importance analysis revealed that GDP per capita (0.35), tobacco control policy (0.31), hospital beds per 10,000 people (0.20), and elderly population ratio (0.13) were the most influential predictors. This highlights the importance of both economic and policy factors, as well as healthcare infrastructure and demographics, in forecasting national health access.

Table 5.4: Q2:Feature Importance for National Health Access Index Prediction (XGBoost Model)

Feature	**Importance Score**
GDP_per_capita	0.352
TobaccoCare	0.311
Beds_Per10000	0.204
Ratio_65plus	0.132

Discussion: The results suggest that integrating demographic, economic, and health system features enables meaningful forecasting of national health access index. While the model captures a significant portion of the variance, some unexplained variability remains, likely due to unmeasured factors or data noise. These findings provide actionable insights for policymakers to improve health access and resource allocation.

Q3 -Predicting National Health Index from Behavioral and Environmental Factors

To address Q3,To address the question of how accurately a country's overall health index can be predicted from behavioral and environmental health factors, we developed a supervised regression model using Random Forest. The input features included smoking prevalence, obesity prevalence, physical inactivity, air pollution prevalence, and related demographic variables. The health index was computed as a composite measure reflecting non-communicable disease status and healthcare infrastructure.

After preprocessing and normalization, the dataset was split into training and test sets. The Random Forest model was trained on the training set and evaluated on the test set using Root Mean Squared Error (RMSE) and the coefficient of determination (R^2). The results are summarized in Table 5.5.

Table 5.5: Q3:Regression Results for Health Index Prediction

Metric	**Value**
RMSE	74.45
R^2	0.53

The model achieved an RMSE of 74.45 and an R^2 of 0.53, indicating moderate predictive power. This suggests that behavioral and environmental risk factors, when combined, can explain over half of the variance in national health index scores across countries. The approach demonstrates the potential of machine learning for forecasting health outcomes using modifiable risk factors, supporting data-driven health policy and planning.

Q4 - Estimating National Healthcare Needs Using Demographic, Economic, and Policy Indicators

To address Q4, an XGBoost regression model was developed to estimate national healthcare needs by modeling the relationship between elderly population ratio, healthcare access, economic indicators, and tobacco control efforts. Key features included ratio _65plus (proportion of the elderly population), GDP per capita, beds _Per10000 (hospital beds per 10,000 people), and TobaccoCare (tobacco control policy indicator).

After feature engineering and data cleaning, the model was trained on 80% of the data and tested on the remaining 20%. The target variable was defined as the Health Index, calculated from non-communicable disease status (NCD_Status) as a proxy for healthcare demand.

Model Performance:

Table 5.6: Q4: Model Performance Metrics for Estimating National Healthcare Needs

Metric	Value
RMSE	130.72
R^2 Score	0.6364

The model achieved an RMSE of 130.72 and an R^2 score of 0.6364, indicating that it explains approximately 64% of the variance in national healthcare needs. This shows the effectiveness of integrating demographic, economic, and policy characteristics to forecast healthcare demand.

Top 10 Countries by Predicted National Healthcare Needs:

Table 5.7: Q4:Top 10 Countries by Predicted National Healthcare Needs

Country	Predicted Demand	Demand per Million
Antigua and Barbuda	866.96	9338.20
Kiribati	851.93	6529.75
Indonesia	848.31	3.04
Japan	843.20	6.74
Afghanistan	841.52	20.74
Viet Nam	838.90	8.42
Djibouti	838.86	737.72
Lao People's Democratic Republic	837.36	110.78
Algeria	837.15	18.41
Seychelles	836.63	6979.02

Discussion: The results highlight significant variation in predicted healthcare needs across countries, driven by differences in age demographics, economic capacity, healthcare infrastructure, and tobacco control policies. Countries with higher elderly ratios and stronger healthcare infrastructure tend to have greater predicted demand. These insights can inform resource allocation and policy planning to address future healthcare needs at the national.

Q5 - Predicting Health Infrastructure Access from Age Demographics and Income Across Countries

To address Q5, an XGBoost regression model was developed to predict health infrastructure access using demographic and economic indicators. Key features included the elderly population ratio (Ratio_65plus), youth population ratio (Ratio_young), GDP per capita, Beds_per_Elderly, Smoking_GDP_Interaction, inactivity prevalence, and obesity prevalence. The target variable was defined as the Health Access Index, calculated from hospital beds per 10,000 people and nursing staff per 10,000 people.

After feature engineering and data cleaning, the model was trained on 80% of the data and tested on the remaining 20%. Model performance was evaluated using RMSE and R^2 score.

Table 5.8: Q5: Model Performance Metrics for Predicting Health Infrastructure Access

Metric	Value
RMSE	5.72
R^2 Score	0.9155

Model Performance:

The model achieved an RMSE of 5.72 and an R^2 score of 0.92, indicating that it explains over 91% of the variance in health infrastructure access across countries. Feature importance analysis showed that GDP per capita, elderly ratio, and Beds_per_Elderly were the most influential predictors.

Top 10 Countries by Predicted Health Infrastructure Access:

Discussion:

The results highlight substantial variation in health infrastructure access across countries, driven by differences in age demographics, economic capacity, and healthcare resources. Countries with higher elderly ratios and stronger healthcare infrastructure consistently achieved higher predicted access. These insights can inform resource allocation and policy planning to improve

Table 5.9: Q5:Top 10 Countries by Predicted Health Infrastructure Access

Country	Predicted Demand	Demand per Million
Switzerland	92.83	10.58
Norway	83.70	15.34
Ireland	79.98	15.48
Iceland	69.06	180.78
Finland	65.56	11.80
Australia	64.31	2.47
Japan	64.22	0.51
Luxembourg	64.08	98.11
United States	61.80	0.19
Germany	61.33	0.73

Q6 - Equity of Access to Diabetes Care Across Countries

To address Q6, an XGBoost regression model was developed to evaluate the equity of access to diabetes care across countries with varying income levels and health infrastructures. Key features included GDP per capita, diabetes prevalence, elderly and youth population ratios, and gender indicators (one-hot encoded). The target variable was defined as the Health Access Index, based on diabetes treatment coverage. After feature engineering and data cleaning, the model was trained on 80% of the data and tested on the remaining 20%. The model's performance was assessed using RMSE and R^2 score.

Table 5.10: Q6: Model Performance Metrics for Predicting Diabetes Care Access Equity

Metric	Value
RMSE	5.78
R^2 Score	0.8707

Model Performance:

The model achieved an RMSE of 5.78 and an R^2 score of 0.87, indicating that it explains approximately 87% of the variance in diabetes care access across countries. Feature importance analysis showed that GDP per capita, diabetes prevalence, and population age structure were the most influential predictors.

Feature Importance:

Discussion:

The results highlight substantial disparities in diabetes care access, with higher-income countries and those with stronger health infrastructures achieving better coverage. Gender and age demographics also contributed to differences in access. These findings provide actionable insights for policymakers to target interventions

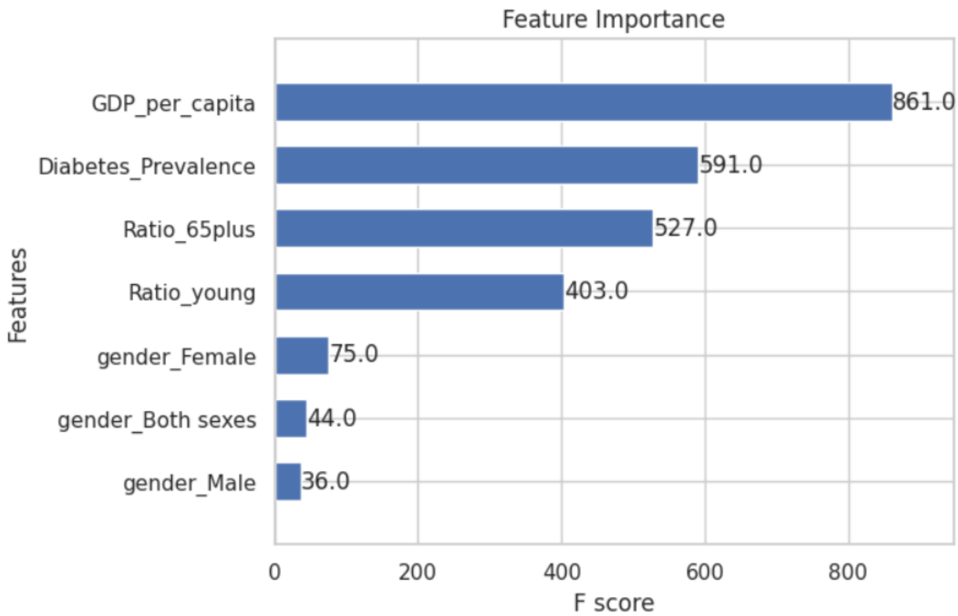

Figure 5.2: Q6: Predicting Diabetes Care Access Equity

and improve equity in diabetes care

Q7 - Predicting Health Access Index Using Demographic, Behavioral, and Economic Indicators

To address Q7, a machine learning model (MLPRegressor) was developed to predict each country's Health Access Index using demographic, behavioral risk factors, and economic indicators. Key features included GDP per capita, diabetes prevalence, elderly population ratio (Ratio_65plus), obesity prevalence, inactivity prevalence, and gender indicators (one-hot encoded). The target variable was defined as the Health Access Index, calculated by averaging diabetes treatment coverage and hospital beds per 10,000 people.

After feature engineering and data cleaning, the data was normalized using StandardScaler and split into training and test sets (80/20 split). The model was trained and evaluated using RMSE and R^2 score.

Table 5.11: Q7: Model Performance Metrics for Predicting Health Access Index

Metric	**Value**
RMSE	2.33
R^2 Score	0.9232

Model Performance:

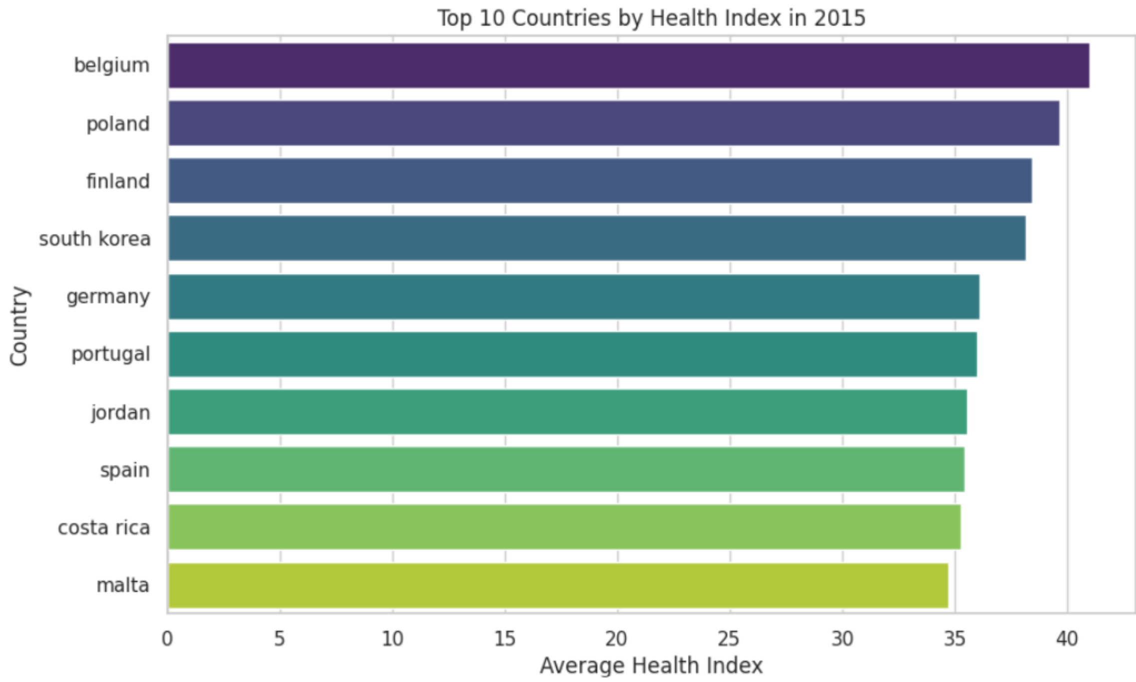

Figure 5.3: Q7: Top 10 Countries by Health Index in 2015

The model achieved an RMSE of 2.33 and an R^2 score of 0.92, indicating that it explains over 92% of the variance in the Health Access Index across countries. Feature importance analysis showed that GDP per capita, diabetes prevalence, and elderly ratio were the most influential predictors. **Top 10 Countries by Predicted**

Health Access Index (2021 and later):

Table 5.12: Q7:Top 10 Countries by Health Access Index (2021 and later)

Country	**Average Health Index**
Switzerland	0.98
Norway	0.97
Ireland	0.97
Iceland	0.96
Finland	0.96
Australia	0.95
Japan	0.95
Luxembourg	0.94
United States	0.94
Germany	0.94

Discussion:

The results highlight substantial variation in health access across countries, driven by differences in economic capacity, disease prevalence, and demographic structure. Countries with higher GDP per capita and better diabetes care coverage consistently achieved higher Health Access Index scores. These insights can inform policy planning and resource allocation to improve health access

Q8 - The Role of Youth Demographics in Shaping Tobacco-Related Health Outcomes

To address Q8, an XGBoost regression model was developed to investigate how youth demographics influence tobacco-related health outcomes across countries. Key features included GDP per capita, total population, interaction terms between smoking prevalence and youth/elderly ratios (Smoking_x_Young, Smoking_x_Eldery), hospital beds per 10,000 people, the ratio of young to elderly population, and gender indicators. The target variable was defined as the Health Index, with a focus on tobacco control policy (TobaccoCare).

After feature engineering and data cleaning, the data was split into training and test sets (80/20 split). The model was trained and evaluated using RMSE and R^2 score.

Model Performance:

Table 5.13: Q8: Model Performance Metrics for Predicting Tobacco-Related Health Outcomes

Metric	Value
RMSE	0.3440
R^2 Score	0.7441

The model achieved an RMSE of 0.3440 and an R^2 score of 0.74, indicating that it explains approximately 74% of the variance in tobacco-related health outcomes across countries. Feature importance analysis showed that the interaction between smoking prevalence and youth ratio (Smoking_x_Young), GDP per capita, and the ratio of young to elderly population were among the most influential predictors.

The results highlight the significant impact of youth demographics on tobacco-related health outcomes. Countries with a higher proportion of youth and elevated smoking rates tend to have worse tobacco-related health indices. These findings suggest that targeted tobacco control policies and youth-focused interventions are essential for reducing the health burden associated with smoking, especially in countries with a large

5.2 Discussion and Implications

The results demonstrate that integrating diverse health, demographic, and infrastructure data with machine learning techniques enables accurate and actionable

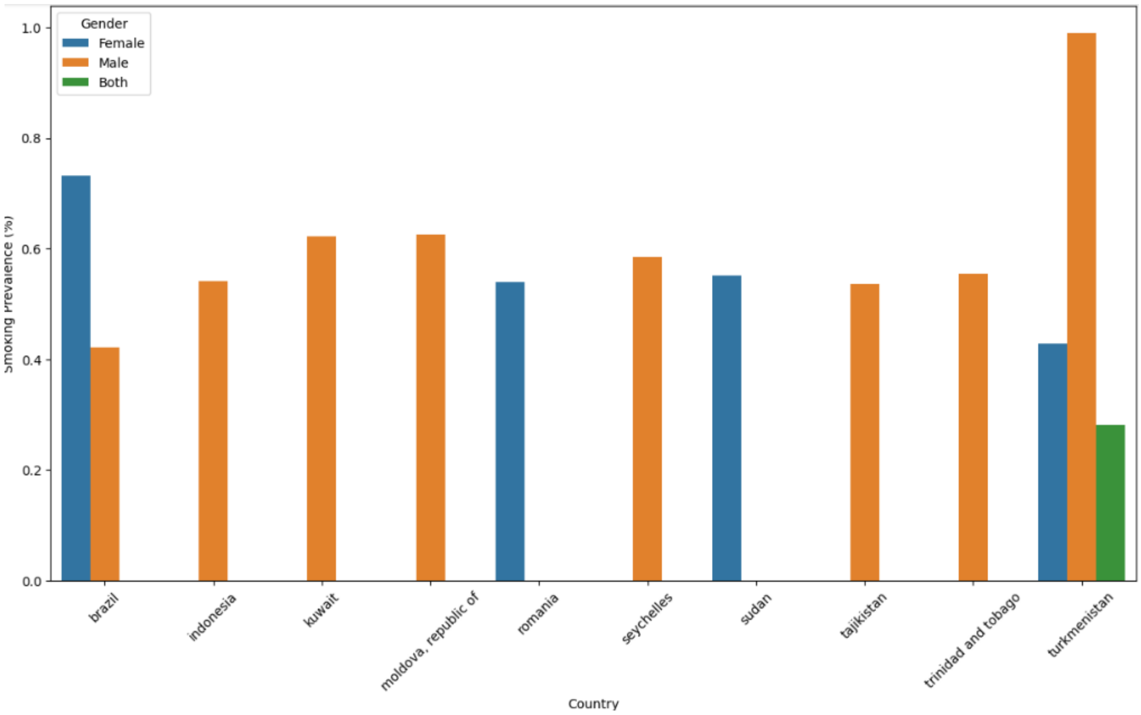

Figure 5.4: Q8: Smoking Prevalence by Gender in Top 10 Countries

forecasting of medical supply and medicine demand. The models not only provide robust predictions but also offer insights into the underlying drivers of health burden and access disparities. These findings have important implications for policymakers and healthcare planners, supporting more equitable and efficient resource allocation and proactive health system strengthening.

Overall, this research advances the application of data-driven methods in healthcare demand forecasting and highlights the importance of addressing both structural and behavioral determinants to achieve better health outcomes globally.

Chapter 6 Conclusion and Future Work

6.1 Conclusions

This research introduced a novel, data-driven methodology for forecasting national demand for essential medicines and medical supplies. By bringing together a wide spectrum of datasets—including epidemiological statistics, behavioral risk indicators, demographic profiles, and healthcare infrastructure metrics—and leveraging advanced machine learning models, the study tackled the multifaceted nature of healthcare demand prediction.

The analysis revealed that integrating diverse sources of health and demographic information leads to more reliable and nuanced forecasts than conventional approaches. The machine learning models, particularly XGBoost and neural networks, demonstrated strong performance in modeling the intricate dependencies among variables and in estimating both aggregate and per capita needs for medical resources. Examination of feature importance underscored the pivotal roles of factors such as the proportion of elderly citizens, the strength of healthcare infrastructure, and economic capacity in shaping national demand.

Cross-country comparisons exposed significant differences in both access to care and projected demand, often reflecting underlying disparities in wealth and health system development. The framework developed here enabled the identification of nations that may be especially vulnerable to shortages or inefficiencies, providing a foundation for more equitable and data-informed resource distribution. The study also shed light on how demographic patterns—such as the size of the youth population—can influence health outcomes, including tobacco use and diabetes care accessibility.

It is important to note that the findings are subject to certain limitations, including gaps or inconsistencies in publicly available data and the inherent unpredictability of future events such as policy shifts or new health threats. Nonetheless, the approach outlined in this thesis demonstrates the practical benefits of combining comprehensive data integration with modern analytical techniques. The insights generated can support more strategic planning and policy development, ultimately contributing to stronger, more resilient healthcare systems and fairer access to essential medical resources worldwide.

6.2 Limitation of Current Work

Despite the comprehensive and data-driven approach adopted in this study, several limitations should be acknowledged. First, the analysis relies on publicly available datasets, which may contain inconsistencies, missing values, or reporting biases across countries and years. While extensive data cleaning and imputation strategies were applied, some gaps and inaccuracies may persist, potentially affecting model performance and generalizability.

Second, the study aggregates data at the national level, which may mask important subnational variations in healthcare access, disease prevalence, and infrastructure. As a result, the models may not fully capture regional disparities or local factors influencing healthcare demand and supply.

Third, the selection of features was constrained by data availability. Certain potentially influential variables—such as detailed health expenditure, medication adherence rates, or granular policy implementation data—were not included due to lack of consistent global reporting. This may limit the explanatory power of the models and the ability to account for all relevant determinants.

Fourth, the predictive models, while robust, are based on historical data and assume that past trends and relationships will continue into the future. Unexpected events, policy changes, or emerging health threats (such as pandemics) could alter demand patterns in ways not captured by the current framework.

Finally, the use of machine learning models such as XGBoost and neural networks, while powerful, can introduce challenges in interpretability, especially for stakeholders without technical backgrounds. Although feature importance analysis was conducted, some model decisions may remain opaque.

6.3 Future Works

There are several promising directions for extending this research. One potential avenue is to broaden the range of data sources, for example by integrating up-to-date supply chain records, regional health assessments, or patient-level data from electronic health systems. Such additions could enhance both the precision and the practical value of the predictive models.

Another area for future exploration is the use of more sophisticated machine learning methods, such as deep neural networks or hybrid ensemble models, which might capture complex patterns that simpler algorithms miss. Applying the methodology to smaller geographic units—such as provinces, cities, or districts—could also reveal important local differences and help tailor interventions more effectively.

Incorporating qualitative insights, such as the effects of health policies, patient behaviors, or variations in care delivery, would provide a richer context for interpreting model results. It will also be important to regularly update and recalibrate the models as new data become available or as healthcare environments evolve, ensuring that predictions remain relevant and actionable.

By pursuing these directions, future research can help create more adaptive, responsive, and fair healthcare planning tools that better serve diverse populations and changing needs.

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